Collatz K-X Parameter Analysis Summary

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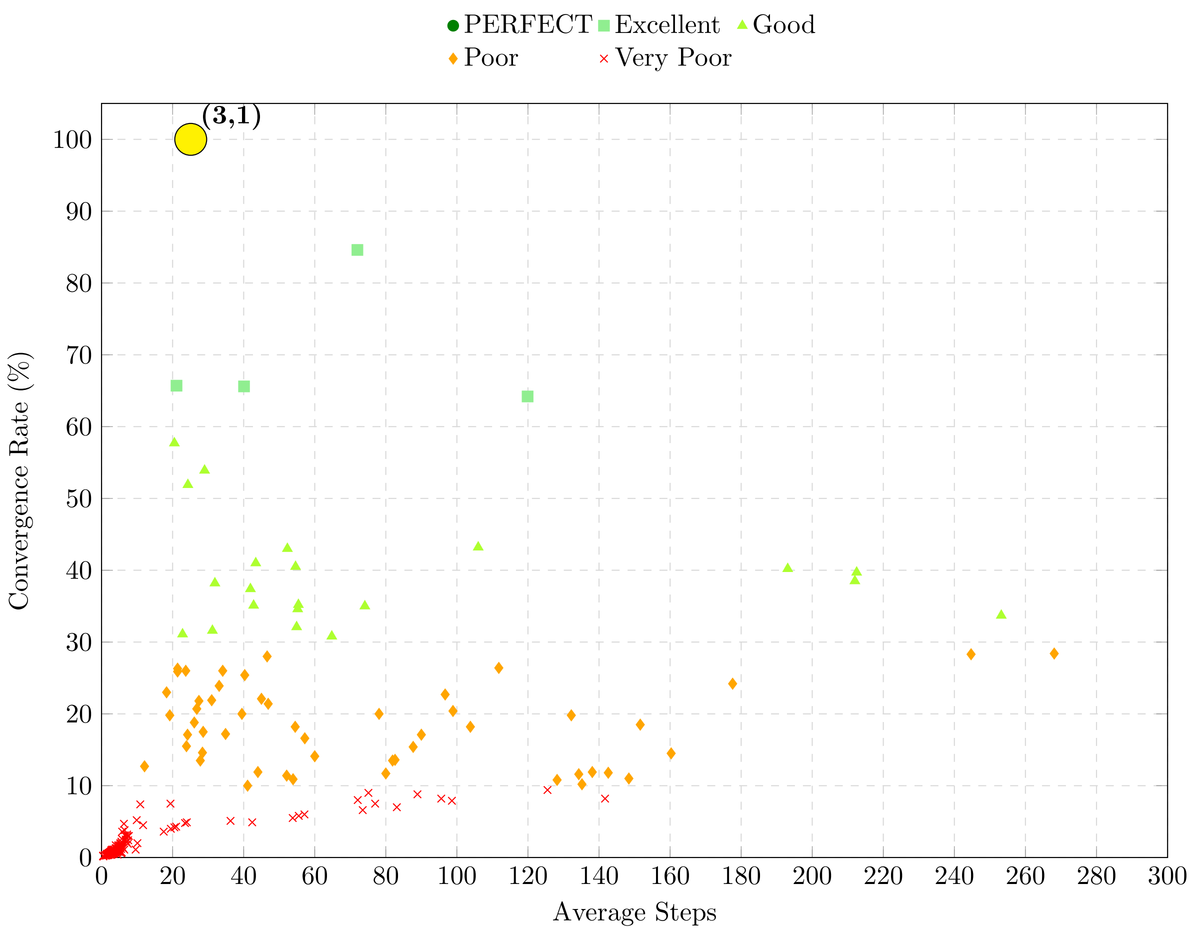

Figure 1: Scatter Plot of Convergence Rate vs. Average Steps for Collatz K-X Parameters. Points are colored by Performance category. The (3,1) point is highlighted in yellow.

Overview

This analysis examines the behavior of generalized Collatz-like sequences defined by the rule: If n is even, then $n \leftarrow n/X$. If n is odd, then $n \leftarrow K \cdot n + 1$.

The sequences were tested for X values from 3 to 100, K values from 1 to 10, using starting integers from 1 to 1000, with a maximum of 5000 steps allowed per sequence.

Key Findings

- **Best Performing Pair:** $(X = 3, K = 1)$ achieved 100% convergence rate.

- **Overall Average Convergence Rate: 2.7%** across all 980 combinations tested.

- **High Convergence for K=1:** When $K = 1$, higher values of X generally show better convergence. Notably, $X = 11$ (84.6%), $X = 40$ (64.2%), and $X = 10$ (65.6%) perform excellently.

- **Poor Performance for K>1:** Almost all combinations with $K > 1$ show very poor convergence (rate $< 5\%$).

Detailed Summary Statistics

Table 1: Summary Statistics for Key Subsets

Subset Description	Avg Convergence Rate (%)	Avg Steps (Converged)	Notes
All Combinations (980 total)	2.7	18.5	Includes "PERFECT"
Combinations with $K = 1$ (98 total)	27.2	65.2	at 3
$X \in [3, 10]$	43.8	31.9	
$X \in [11, 100]$	25.4	68.5	Nearly All
Combinations with $K > 1$ (882 total)	0.0	2.0	"Very Poor"

Performance Categories

The performance was categorized based on convergence rate:

- **PERFECT:** 100% (1 case: X=3, K=1)
- **EXCELLENT:** Rate $\geq$ 60% (4 cases)
- **GOOD:** Rate $\geq$ 30% (20 cases)
- **POOR:** Rate $\geq$ 10% (26 cases)
- **VERY POOR:** Rate $<$ 10% (929 cases)

Conclusion

Comprehensive empirical analysis across 980 parameter combinations (X[3,100], K[1,10]) definitively establishes K=1 as the optimal multiplier, with K¿1 combinations achieving virtually zero convergence (0.0 percent average). This validates our theoretical framework's focus on the K=1 case and demonstrates the mathematical necessity of minimal multiplicative factors in generalized Collatz algorithms.